

Name: _____

Teacher: _____



YEAR 12 MATHEMATICS

EXTENSION 1

HALF YEARLY EXAMINATION 2005

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working must be shown in every question
- All questions are of equal value
- Start each question on a separate page

Question 1**Marks**

- (a) Find the size of the acute angle between the lines 3

$$y = -x$$

$$\sqrt{3}y = x$$

- (b) Solve the inequality $\frac{2x+1}{x-1} > 3$. 2

- (c) A sector of angle 135° at the centre, is cut from a circular piece of cardboard of radius 8cm. The cut edges are brought together to form a cone. Find the circumference of the base of the cone. 2

- (d) For the given function $f(x) = \frac{8}{4+x^2}$,

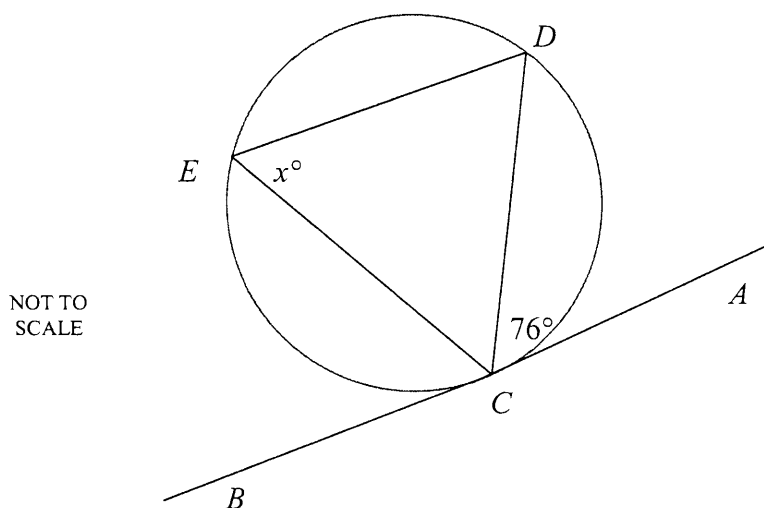
- (i) show that $f(x)$ is an even function. 2

- (ii) evaluate $\lim_{x \rightarrow \infty} \frac{8}{4+x^2}$. 2

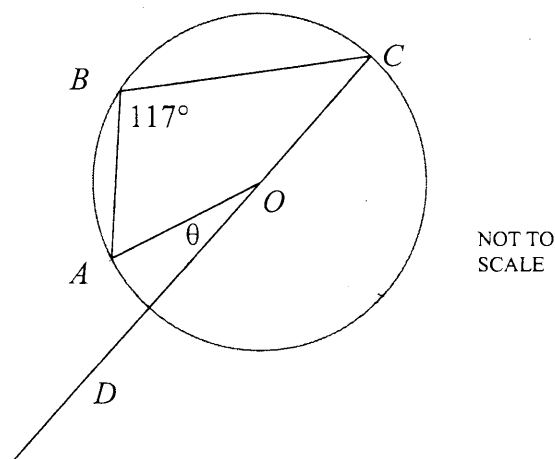
- (iii) sketch the graph of $y = f(x)$. 1

Question 2*Start a new page***Marks**

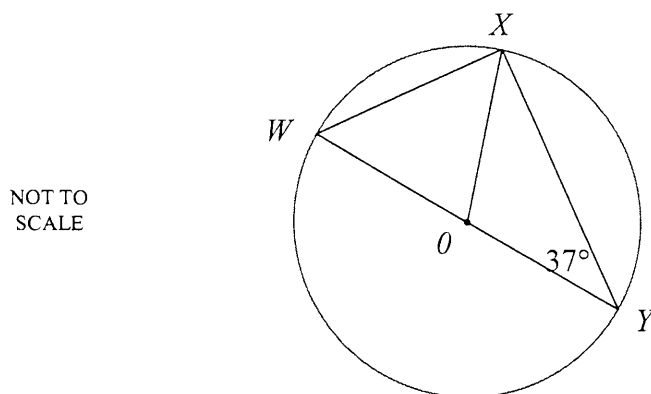
- (a) In the diagram drawn below AB is a tangent to the circle and $\angle DCA = 76^\circ$.
Find the value of the pronumeral, giving reasons for your answer.

2

- (b) Given that O is the centre of the circle and $\angle ABC = 117^\circ$,
find the value of θ , giving reasons for your answer.

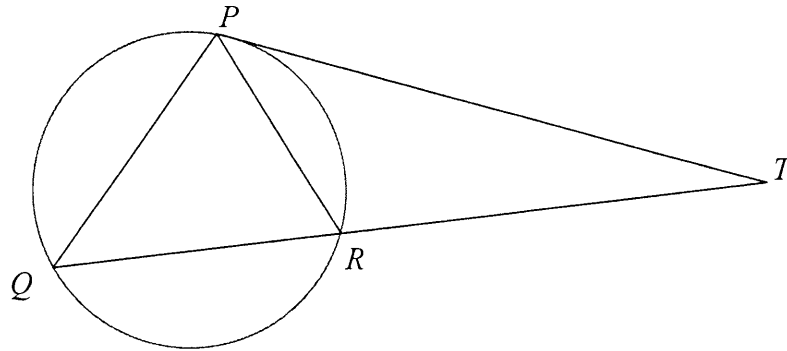
2

- (c) In the diagram drawn below, WY is a diameter of a circle, centre O .
If $\angle WYX = 37^\circ$, find the size of $\angle WXO$.
Give reasons for your answer.

2

Question 2 (cont.)**Marks**

- (d) PT is a tangent to the circle drawn below and QR is a secant, intersecting the circle at Q and at R . The line QR intersects PT at T .



- (i) Prove that the triangles PRT and QPT are similar.

2

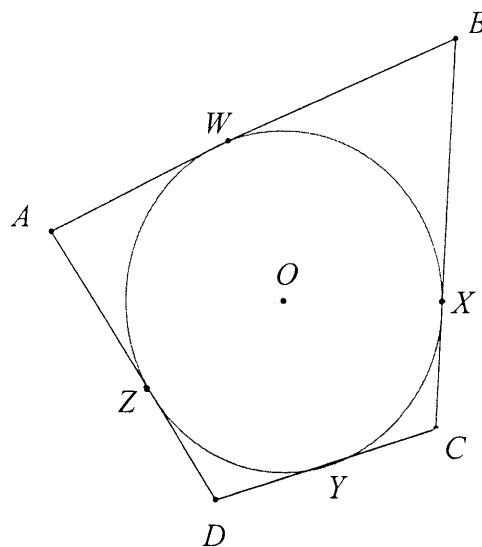
- (ii) Hence prove $PT^2 = QT \times RT$.

1

- (e) A quadrilateral $ABCD$ is constructed so that AB , BC , CD and DA are tangents to a circle. W , X , Y and Z are the points of contact of the tangents AB , BC , CD and DA respectively.

- (i) Copy the diagram neatly.

- (ii) Prove that $AB + DC = AD + BC$.

3

Question 3*Start a new page***Marks**

- (a) Find the value of k for which $(x+2)$ is a factor of the polynomial

3

$$2x^3 + kx^2 - 18x - 8.$$

Hence, express the polynomial as a product of its linear factors.

- (b) Sketch the graph of the polynomial

2

$$P(x) = x(x+1)^2(2x-1).$$

- (c) If α, β, γ are the roots of the equation $x^3 + 2x^2 + 3x + 4 = 0$,

3

find the value of (i) $\alpha + \beta + \gamma$

(ii) $\alpha\beta + \alpha\gamma + \beta\gamma$

(iii) $\alpha^2 + \beta^2 + \gamma^2$.

- (d) The polynomial $x^3 - 6x^2 + 9x - k$ has a double root.

4

Show that there are two possible values of k .

Find the roots for each value of k .

Question 4*Start a new page***Marks**

(a) Sketch the function $y = 2 \sin 3x$ for $0 \leq x \leq 2\pi$.

2

(b) Differentiate $\cos^4 x$.

2

(c) Solve $7 \sin \theta + \cos \theta = 5$ for $0^\circ \leq \theta \leq 360^\circ$.

3

Write your solution(s) to the nearest minute.

(d) Use the table of standard integrals to show that:

2

$$\int_0^{\frac{\pi}{9}} \sec 3x \tan 3x \, dx = \frac{1}{3}$$

(e) Prove that $\frac{1}{\tan A + \cot B} + \frac{1}{\cot A + \tan B} = \frac{\sin(A+B)}{\cos(A-B)}$.

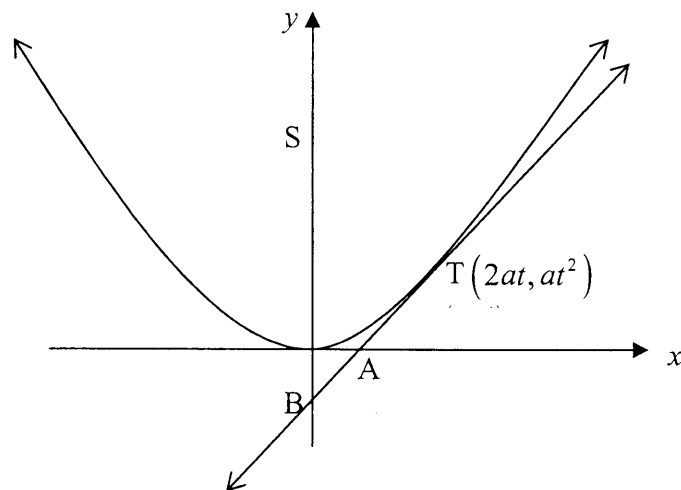
3

Question 5*Start a new page***Marks**

- (a) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$, are two points on the parabola $x^2 = 4ay$. If PQ is a focal chord, prove that $pq = -1$.

3

- (b) $T(2at, at^2)$ is a variable point on the parabola $x^2 = 4ay$.



- (i) Show that the equation of the tangent, to the parabola at the point T, is $y - tx + at^2 = 0$.
- (ii) If the tangent at T cuts the x -axis at A, and the y -axis at B, find the co-ordinates of A and B.
- (iii) Show that the tangent at T makes equal angles with the y -axis and the line TS, where S is the focus of the parabola.
- (iv) In what ratio does the point T, divide the interval AB?

1**2****3****3**

Question 6*Start a new page***Marks**

(a) Solve $\log_3(9x - 2) - 2\log_3 x = 2$. 3

(b) Evaluate $\int_0^1 \frac{3x^2}{1+x^3} dx$. 2

(c) Calculate the volume of the solid of revolution, 2

formed by rotating the curve $y = e^x + e^{-x}$, about the x -axis,

between $x = -1$ and $x = 1$.

(d) Consider the function $y = xe^{-x}$.

(i) Determine the nature of any stationary point(s). 2

(ii) Find any point(s) of inflexion. 2

(iii) Sketch the function. 1

Question 7*Start a new page***Marks**

(a) Evaluate $\int_0^4 x\sqrt{16-x^2} dx$,

3

using the substitution $u = 16 - x^2$, or otherwise.

- (b) A solid of revolution is formed by rotating the area under the curve $y = \tan x$ between $x = 0$ and $x = \frac{\pi}{3}$, around the x -axis.

3

Find the exact volume of the solid.

(c) Show that $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$.

3

Hence, find the exact value of $\int_0^{\frac{\pi}{2}} \sin^2 \frac{\theta}{2} d\theta$.

- (d) For a certain function, $f''(x) = -18 \cos 3x$.

3

Determine the equation of this function,

given that there is a stationary point at the point $\left(\frac{2\pi}{3}, 1\right)$.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

QUESTION 1.

(a) $y = -x$

$m_1 = -1$

$y = \frac{1}{\sqrt{3}}x$

$m_2 = \frac{1}{\sqrt{3}}$

ie/ $m_2 = \sqrt{3}/3$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-1 - \sqrt{3}/3}{1 - \sqrt{3}/3} \right|$$

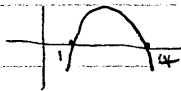
$$= \frac{3 + \sqrt{3}}{3 - \sqrt{3}}$$

$\therefore \theta = 75^\circ$

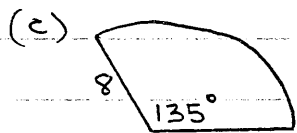
(b) $(2x+1)(x-1) > 3(x-1)^2$

$(x-1)[2x+1-3(x-1)] > 0$

$(x-1)(4-x) > 0$



$\therefore 1 < x < 4$



$l = r\theta$

$= 8 \times \frac{3\pi}{4}$

$= 6\pi$

$\therefore$ the circumf. is 6π cm.

(d) (i) $f(-x) = \frac{8}{4+(-x)^2}$

$= \frac{8}{4+x^2}$

$= f(x)$

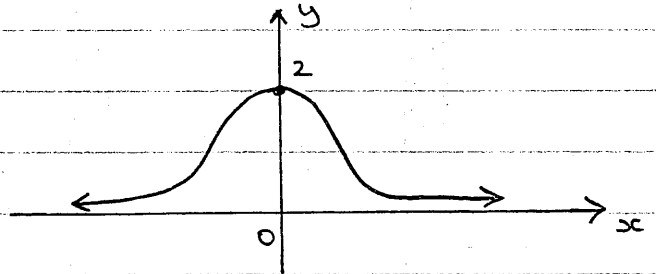
$\therefore f(x)$ is an even fn.

(ii) $\lim_{x \rightarrow \infty} \frac{8}{4+x^2} = \lim_{x \rightarrow \infty} \frac{8/x^2}{4/x^2+1}$

$= \frac{0}{0+1}$

$= 0$

(iii) when $x=0$, $y=2$



QUESTION 2.

a) $x = 76$ (alt. seg. thm)

b) reflex $\angle AOC = 234^\circ$ (angle at centre is twice $\angle$ at circumf.)

$$\theta = 234^\circ - 180^\circ \text{ (COB is a diam)}$$

$$\therefore \theta = 54^\circ$$

c) $\angle WXY = 90^\circ$ ($\angle$ in a semi-circle)

$$\angle OXY = 37^\circ \text{ (base } \angle \text{'s of isos. } \Delta \text{)}$$

$$\therefore \angle WXD = 53^\circ$$

d) i) In $\triangle PRT$ and $\triangle QPT$

LT is common

$$\angle TPR = \angle TPQ \text{ (alt. seg. thm)}$$

$$\therefore \triangle PRT \parallel \triangle QPT \text{ (equiangular)}$$

ii) $\frac{PT}{QT} = \frac{RT}{PT}$ (ratio of corres. sides in sim. Δ 's)

$$\therefore PT^2 = QT \times RT$$

e) ii) $BW = BX$ (tangents to a circle, from ext. pt., are equal)

$$AW = AZ$$

"

"

"

"

$$DY = DZ$$

"

"

"

"

$$CY = CX$$

"

"

"

"

$$\text{So, } (AW + BW) + (DY + CY) = (AZ + BX) + (DZ + CX)$$

$$\text{ie } AB + DC = (AZ + DZ) + (BX + CX)$$

$$= AD + BC$$

$$\therefore AB + DC = AD + BC$$

QUESTION 3

(a) Let $P(x) = 2x^3 + Kx^2 - 18x - 8$, then $P(-2) = 0$

$$0 = -16 + 4K + 36 - 8$$

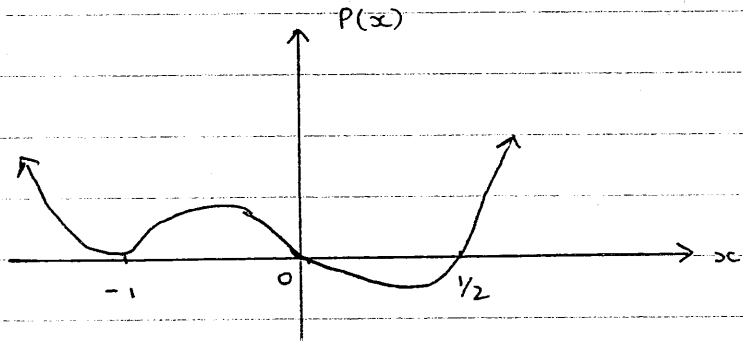
$$\therefore K = -3$$

$$\begin{array}{r} 2x^2 - 7x - 4 \\ x+2 \overline{) 2x^3 - 3x^2 - 18x - 8} \\ \underline{2x^3 + 4x^2} \\ -7x^2 - 18x \\ \underline{-7x^2 - 14x} \\ -4x - 8 \\ \underline{-4x - 8} \end{array}$$

$$\therefore 2x^3 - 3x^2 - 18x - 8 = (x+2)(x-4)(2x+1)$$

(b) $P(x) = x(x+1)^2(2x-1)$

$$P(1) > 0$$



(c) (i) $\alpha + \beta + \gamma = -b/a$
 $= -2$

(ii) $\alpha\beta + \alpha\gamma + \beta\gamma = c/a$
 $= 3$

(iii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$
 $= 4 - 6$
 $= -2$

(d) Let the roots be α, α, β

then $2\alpha + \beta = 6$ — (1)

$$\alpha^2 + 2\beta\alpha = 9$$
 — (2)

$$\alpha^2\beta = K$$
 — (3)

from ① $\beta = 6 - 2\alpha$

sub. in ② $\alpha^2 + 2\alpha(6 - 2\alpha) = 9$

$$\alpha^2 + 12\alpha - 4\alpha^2 = 9$$

$$-3\alpha^2 + 12\alpha - 9 = 0$$

$$\alpha^2 - 4\alpha + 3 = 0$$

$$(\alpha - 3)(\alpha - 1) = 0$$

$$\alpha = 3, 1$$

when $\alpha = 3$, $\beta = 0$ and $K = 0$

" $\alpha = 1$, $\beta = 4$ and $K = 4$

$\therefore$ if $K = 0$, the roots are 3, 3, 0

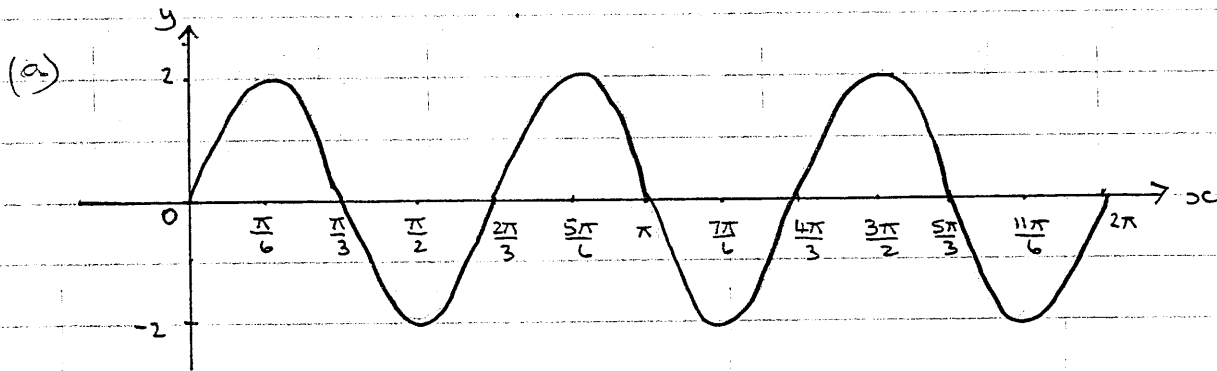
if $K = 4$, " " " 1, 1, 4

QUESTION 4.

$$\begin{aligned} \text{e) } \tan A + \cot B &= \frac{\sin A}{\cos A} + \frac{\cos B}{\sin B} \\ &= \frac{\sin A \sin B + \cos A \cos B}{\cos A \sin B} \\ &= \frac{\cos(A - B)}{\cos A \sin B} \end{aligned}$$

$$\begin{aligned} \cot A + \tan B &= \frac{\cos A}{\sin A} + \frac{\sin B}{\cos B} \\ &= \frac{\cos A \cos B + \sin A \sin B}{\sin A \cos B} \\ &= \frac{\cos(A - B)}{\sin A \cos B} \end{aligned}$$

$$\begin{aligned} \text{LHS} &= \frac{\cos A \sin B}{\cos(A - B)} + \frac{\sin A \cos B}{\cos(A - B)} \\ &= \frac{\cos A \sin B + \sin A \cos B}{\cos(A - B)} \\ &= \frac{\sin(A + B)}{\cos(A - B)} \\ &= \text{RHS.} \end{aligned}$$



(b) Let $y = \cos^4 x$

$$\frac{dy}{dx} = 4 \cos^3 x \times -\sin x$$

ie/ $\frac{dy}{dx} = -4 \sin x \cos^3 x$

(c) $7 \sin \theta + \cos \theta = R \sin(\theta + \alpha)$

$$R = \sqrt{50}$$

ie $R = 5\sqrt{2}$

$$\tan \alpha = 1/7$$

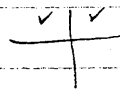
$$\alpha = 8^\circ 8'$$

$$5\sqrt{2} \sin(\theta + 8^\circ 8') = 5$$

$$0^\circ \leq \theta \leq 360^\circ$$

$$\sin(\theta + 8^\circ 8') = 1/\sqrt{2}$$

$$8^\circ 8' \leq \theta \leq 368^\circ 8'$$



$$\theta + 8^\circ 8' = 45^\circ \text{ or } 135^\circ$$

$$\theta = 36^\circ 52' \text{ or } 126^\circ 52'$$

or// $7 \cdot \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = 5$

$$0^\circ \leq \frac{\theta}{2} \leq 180^\circ$$

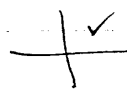
$$14t + 1 - t^2 = 5 + 5t^2$$

$$6t^2 - 14t + 4 = 0$$

$$3t^2 - 7t + 2 = 0$$

$$(3t-1)(t-2) = 0$$

$$\tan \theta/2 = 1/3 \text{ or } 2$$



$$\theta/2 = 18^\circ 26' \text{ or } 63^\circ 26'$$

$$\therefore \theta = 36^\circ 52' \text{ or } 126^\circ 52'$$

$$\begin{aligned}
 (d) \int_0^{\pi/9} \sec 3x \tan 3x \, dx &= \frac{1}{3} \left[\sec 3x \right]_0^{\pi/9} \\
 &= \frac{1}{3} \left[\sec \pi/3 - \sec 0 \right] \\
 &= \frac{1}{3} (2 - 1) \\
 &= \frac{1}{3}
 \end{aligned}$$

QUESTION 5

$$\begin{aligned}
 (a) \text{ Eqn of chord PQ: } m &= \frac{ap^2 - aq^2}{2ap - 2aq} \\
 &= \frac{a(p-q)(p+q)}{2a(p-q)} \\
 &= \frac{p+q}{2}
 \end{aligned}$$

$$y - ap^2 = \frac{p+q}{2} (x - 2ap)$$

Since PQ is a focal chord, it passes through the focus $(0, a)$

$$a - ap^2 = \frac{p+q}{2} (0 - 2ap)$$

$$a - ap^2 = -ap(p+q)$$

$$a - \cancel{ap^2} = -\cancel{ap^2} - apq$$

$$a = -apq$$

$$\therefore pq = -1$$

(b) i) $y = x^2 / 4a$

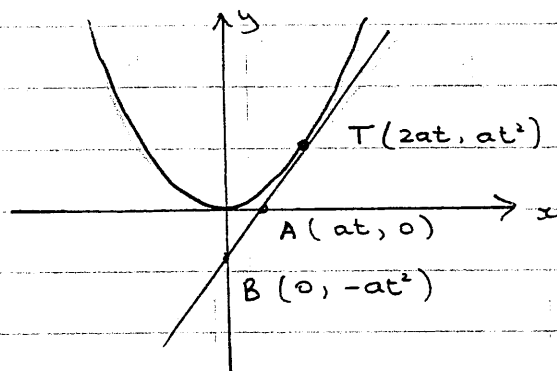
$$\frac{dy}{dx} = x / 2a$$

at T, $m = 2at / 2a$
 $= t$

$$y - at^2 = t(x - 2at)$$

$$y - at^2 = tx - 2at^2$$

$\therefore y - tx + at^2 = 0$, is the eqⁿ of the tangent at T.



ii) when $y = 0$, $x = at \therefore A$ is the point $(at, 0)$

when $x = 0$, $y = -at^2 \therefore B$ is the point $(0, -at^2)$

iii) Show that ΔSTB is isosceles

$$SB = a + at^2$$

$$= a(1 + t^2)$$

$$ST^2 = (2at - 0)^2 + (at^2 - a)^2$$

$$= 4a^2t^2 + a^2t^4 - 2a^2t^2 + a^2$$

$$= a^2t^4 + 2a^2t^2 + a^2$$

$$= a^2(t^4 + 2t^2 + 1)$$

$$= a^2(t^2 + 1)^2$$

$$ST = a(t^2 + 1)$$

$$\therefore SB = ST$$

$\therefore$ the tangent is equally inclined to the y-axis and the focal chord through T.

(iv) $A(x_1, y_1) = (at, 0)$ $B(x_2, y_2) = (0, -at^2)$ $k : l$ $T(x, y) = (2at, at^2)$

$$2at = \frac{k \cdot 0 + l \cdot at}{k + l}$$

$$at^2 = \frac{-kat^2 + l \cdot 0}{k + l}$$

* or, use
the diag.

$$k + l = \frac{lat}{2at}$$

$$= \frac{l}{2}$$

$$k + l = \frac{-kat^2}{at^2}$$

$$= -k$$

$$\therefore \frac{l}{2} = -k$$

$$\therefore \frac{k}{l} = -\frac{1}{2}$$

$\therefore T$ divides AB externally
in the ratio $1 : 2$

QUESTION 6.

$$a) \log_3 \left(\frac{9x-2}{x^2} \right) = 2$$

$$\therefore \frac{9x-2}{x^2} = 3^2$$

$$9x-2 = 9x^2$$

$$9x^2 - 9x + 2 = 0$$

$$(3x-2)(3x-1) = 0$$

$$\therefore x = 2/3, 1/3$$

$$(b) \int_0^1 \frac{3x^2}{1+x^3} dx = \left[\log(1+x^3) \right]_0^1$$

$$= \log 2 - \log 1$$

$$= \log 2$$

$$(c) V = 2\pi \int_0^1 (e^{2x} + 2 + e^{-2x}) dx$$

$$= 2\pi \left[\frac{e^{2x}}{2} + 2x - \frac{e^{-2x}}{2} \right]_0^1$$

$$= 2\pi \left[\frac{e^2}{2} + 2 - \frac{e^{-2}}{2} - \left(\frac{1}{2} - \frac{1}{2} \right) \right]$$

$$y = e^x + e^{-x}$$

$$y^2 = e^{2x} + 2 + e^{-2x}$$

$$\therefore \text{vol. is } 4\pi (e^2 + 4 - 1/e^2) \text{ cub. units.}$$

$$(d) y = xe^{-x}$$

$$u = x$$

$$v = e^{-x}$$

$$u' = 1$$

$$v' = -e^{-x}$$

$$\frac{dy}{dx} = e^{-x} - xe^{-x}$$

$$\frac{d^2y}{dx^2} = -e^{-x} - (e^{-x} - xe^{-x})$$

$$= -e^{-x} - e^{-x} + xe^{-x}$$

$$= e^{-x} (x - 2)$$

St. pts. occur when $y' = 0$

$$\text{i.e. } e^{-x} (1 - x) = 0$$

$$\therefore x = 1$$

when $x=1$, $y = e^{-1}$, $y'' < 0$

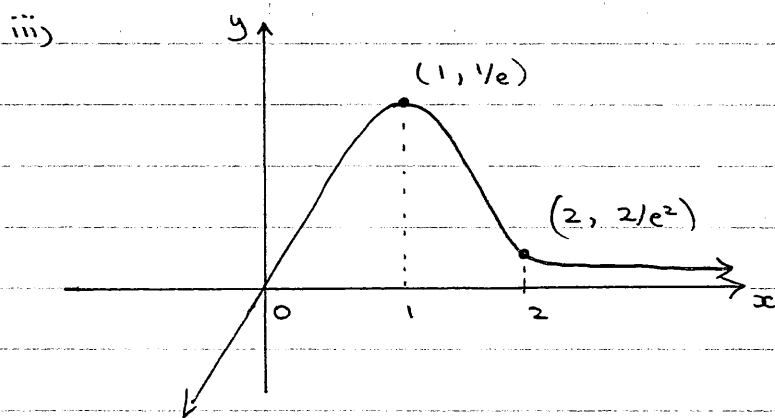
$\therefore$ there is a max. turning point at $(1, 1/e)$

ii) Inflexion occurs when $y'' = 0$ and concavity changes.

$$y'' = 0 \quad \text{when } x=2, \quad y = 2e^{-2}$$

when $x=1$, $y'' < 0$
" $x=3$, $y'' > 0$ } $\therefore$ concavity has changed.

$(2, 2/e^2)$ is a point of inflexion



QUESTION 7

$$\begin{aligned} \text{a) } \int_0^4 x(16-x^2)^{\frac{1}{2}} dx &= -\frac{1}{2} \int_0^4 -2x(16-x^2)^{\frac{1}{2}} dx \\ &= -\frac{1}{2} \cdot \frac{2}{3} \left[(16-x^2)^{\frac{3}{2}} \right]_0^4 \\ &= -\frac{1}{3} \left[0 - 16^{\frac{3}{2}} \right] \\ &= 64/3 \end{aligned}$$

$$\begin{aligned} \text{OR // } \int_0^4 (16-x^2)^{\frac{1}{2}} \cdot x dx &= -\frac{1}{2} \int_{16}^0 u^{\frac{1}{2}} du & u &= 16-x^2 \\ &= -\frac{1}{2} \cdot \frac{2}{3} \left[u^{\frac{3}{2}} \right]_{16}^0 & du &= -2x dx \\ &= -\frac{1}{3} \left[0 - 16^{\frac{3}{2}} \right] & x=0, u &= 16 \\ &= 64/3 & x=4, u &= 0 \end{aligned}$$

$$\begin{aligned}
 \text{b) } V &= \pi \int_0^{\pi/3} \tan^2 x \, dx \\
 &= \pi \int_0^{\pi/3} (\sec^2 x - 1) \, dx \\
 &= \pi \left[\tan x - x \right]_0^{\pi/3}
 \end{aligned}$$

$$\sin^2 x + \cos^2 x = 1$$

$$\tan^2 x + 1 = \sec^2 x$$

$$\begin{aligned}
 &= \pi \left[\tan \pi/3 - \pi/3 - (\tan 0 - 0) \right] \\
 &= \pi \left(\sqrt{3} - \pi/3 \right)
 \end{aligned}$$

$\therefore$ volume is $\pi/3(3\sqrt{3} - \pi)$ cub. units.

$$\text{c) } \cos 2x = \cos^2 x - \sin^2 x$$

$$\cos 2x = 1 - 2\sin^2 x$$

$$2\sin^2 x = 1 - \cos 2x$$

$$\text{ie/ } \sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

$$\begin{aligned}
 \int_0^{\pi/2} \sin^2 \frac{\theta}{2} \, d\theta &= \frac{1}{2} \int_0^{\pi/2} (1 - \cos \theta) \, d\theta \\
 &= \frac{1}{2} \left[\theta - \sin \theta \right]_0^{\pi/2} \\
 &= \frac{1}{2} \left[\frac{\pi}{2} - \sin \frac{\pi}{2} - (0 - \sin 0) \right] \\
 &= \frac{1}{2} \left[\frac{\pi}{2} - 1 \right] \\
 &= \frac{1}{4} (\pi - 2)
 \end{aligned}$$

$$\text{d) } f''(x) = -18 \cos 3x$$

$$f'(x) = -6 \sin 3x + c$$

$$f'(x) = 0 \text{ when } x = \frac{2\pi}{3}$$

$$0 = -6 \sin 2\pi + c$$

$$\text{ie/ } c = 0$$

$$f(x) = 2 \cos 3x + k$$

$$1 = 2 \cos 2\pi + k$$

$$\text{ie/ } k = -1$$

$$\therefore f(x) = 2 \cos 3x - 1$$